

Notes: Cloyne, Ferreira & Surico (RES, 2020)

Monetary Policy when Households have Debt: New Evidence on the Transmission Mechanism

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1 Big picture

- **Question.** How does conventional monetary policy transmit to household consumption when households differ in debt, housing tenure, liquid wealth, and illiquid wealth?
- **Main empirical setting.** The paper studies the U.K. and U.S. using household expenditure surveys and identified monetary policy shocks.
- **Main result.** A 25 bp unanticipated interest-rate cut raises consumption mainly through households with mortgages. Outright owners barely change spending; renters also respond, but less than mortgagors.
- **Mechanism.** Income rises for all groups, but mortgagors and renters have higher MPCs because of liquidity constraints. Many mortgagors are wealthy hand-to-mouth.
- **Economic takeaway.** Monetary transmission works through general-equilibrium income effects coupled with balance-sheet-driven MPC heterogeneity.

2 Data and measurement

2.1 Household expenditure and income surveys

The U.K. analysis uses FES/EFS/LCFS survey data from 1975–2007. The U.S. analysis uses CEX data from 1981–2007. The surveys measure expenditure and income but not high-frequency household balance sheets. Housing tenure is therefore used as a balance-sheet proxy.

Interpretation. The paper turns repeated cross sections into tenure-specific pseudo-cohort time series.

2.2 Housing tenure groups

The three groups are mortgagors, outright owners, and renters. Mortgagors have debt and illiquid housing wealth; outright owners have housing wealth and more liquid wealth; renters have little wealth.

Interpretation. The groups map naturally to high-MPC versus low-MPC theoretical types.

2.3 Monetary policy shocks

The U.K. shocks are from Cloyne and Huertgen; U.S. shocks are an updated Romer–Romer series. Monthly shocks are aggregated to quarterly frequency.

Interpretation. The shock series addresses the usual reverse-causality problem in monetary policy estimation.

2.4 Wealth data

The mechanism evidence uses the BHPS for the U.K. and the SCF for the U.S. These data document net financial wealth, net housing wealth, and wealthy hand-to-mouth status.

Interpretation. Wealth surveys validate the tenure proxy.

3 Models and empirical specifications

3.1 Distributed-lag specification

$$X_{i,t} = \alpha_{0i} + \alpha_{1i}trend + B_i(L)X_{i,t-1} + C_i(L)S_{t-1} + \sum_{q=2}^4 D_q^i Z_q + u_{i,t}.$$

$X_{i,t}$ is the cohort-level outcome; S_t is the monetary policy shock; i indexes mortgagors, outright owners, and renters.

Interpretation. The lag polynomial $C_i(L)$ traces the dynamic impulse response of each housing-tenure group.

3.2 Dollar cumulative response

Dollar change over four years = cumulative IRF $\times$ average group-level dollar value.

Interpretation. This converts percentage responses into comparable dollar changes in expenditure, income, and payments.

3.3 Propensity-score robustness

$$\Pr(Mortgagor_{h,t} = 1 \mid X_{h,t}) = \Phi(\beta' X_{h,t}).$$

The authors classify likely and unlikely mortgagors based on demographics.

Interpretation. This checks that endogenous housing-tenure transitions are not driving the result.

3.4 Wealthy hand-to-mouth definition

$$WHTM = 1\{\text{net illiquid wealth} > 0, \text{net liquid wealth} < 0.5 \times \text{monthly labor income}\}.$$

Interpretation. This captures households with high total wealth but little spendable liquidity.

4 Identification and empirical design

The identification combines external monetary policy shocks, tenure-specific pseudo-cohorts, slow-moving tenure shares, cohort-specific Granger-causality checks, demographic robustness, propensity-score robustness, and mechanism tests using income, payments, wealth, inflation, investment, government spending, and net exports.

5 Tables and figures

5.1 Figure 1: Tenure shares and Figure 2: Monetary policy shocks series

Read: Figure 1 shows slow-moving tenure shares. Figure 2 shows high-frequency policy shocks. The contrast supports the pseudo-cohort design.

Economic message. Housing tenure is slow moving, while monetary shocks supply high-frequency variation.

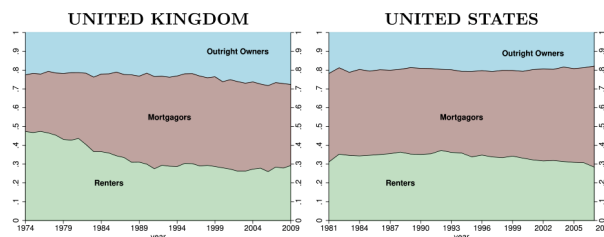


FIGURE 1
Share of mortgage, outright home owners and renters in the U.K. (source: FES/LCFS, 1975–2009) and the U.S. (source: CEX, 1981–2009).

(a) Figure 1: Tenure shares

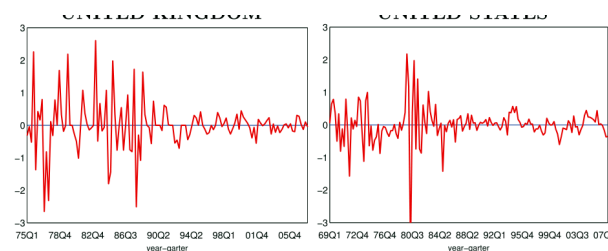


FIGURE 2
Monetary policy shocks series. U.K.: Clovine and Huertgen (2016); U.S.: updated version of Romer and Romer (200)

(b) Figure 2: Policy shocks

5.2 Figure 3: Non-durable consumption by tenure and Figure 4: Durable expenditure by tenure

Read: Mortgagees raise both non-durables and durables. Outright owners respond little. Renters respond, especially for non-durables, but less consistently than mortgagees.

Economic message. The aggregate consumption response is concentrated among mortgagees, especially for durables.

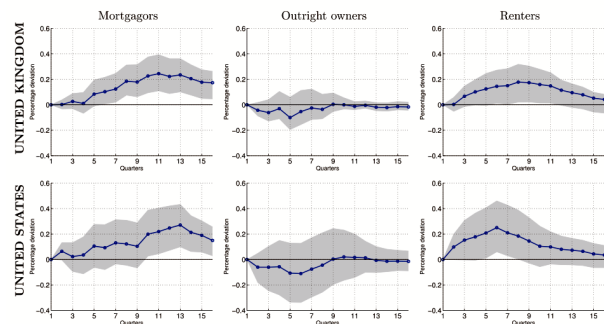


Figure 3: Non-durable goods and services by housing tenure

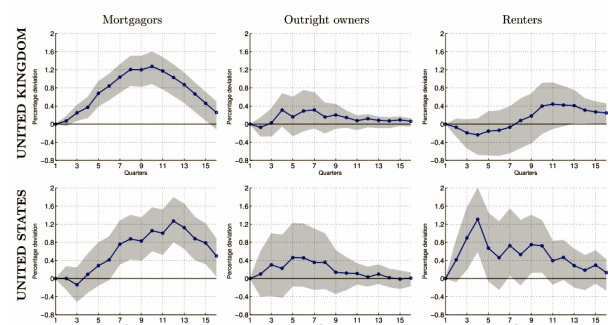


FIGURE 4

Figure 4: Durable expenditure by housing tenure

5.3 Table 1: Cumulative changes over four years in US dollars

Read: Mortgagees' expenditure rises by about \$600 in the U.K. and \$535 in the U.S.; outright owners' spending changes are small; income rises for all groups.

Economic message. Direct mortgage-payment windfalls are too small to explain the spending response.

TABLE 1
Cumulative changes over four years in US\$

Panel A: U.K.				
	Non-durable consumption	Durable expenditure	Mortgage or rental payments	After-tax income
Mortgagors	308.3 [112.8, 516.1]	292.3 [189.2, 369.0]	-166.4 [-272.2, -41.7]	695.9 [186.5, 1105.1]
Outright owners	-62.6 [-148.2, 77.4]	46.5 [-24.6, 107.6]		451.7 [122.5, 797.2]
Renters	155.3 [17.9, 261.8]	19.0 [-36.5, 62.9]	64.7 [4.4, 118.7]	397.3 [94.2, 596.1]

Panel B: U.S.				
	Non-durable consumption	Durable expenditure	Mortgage or rental payments	After-tax income
Mortgagors	305.8 [58.3, 554.3]	229.3 [122.0, 350.8]	-56.3 [-112.8, -4.3]	757.3 [196.8, 1302.0]
Outright owners	-72.3 [-324.8, 186.0]	54.8 [-10.5, 127.8]		585.3 [83.3, 1012.8]
Renters	223.3 [32.3, 412.3]	123.5 [30.3, 213.8]	64.8 [9.8, 121.5]	439.3 [112.8, 699.8]

Notes: The table reports the overall dollar change in expenditure over the four-year period following a temporary 25 bp cut in monetary policy, together with the bootstrapped 90% confidence bands. The magnitudes are per household average. To compute these numbers we cumulate the IRFs and divide by the average value of each variable for each cohort in each country (shown in [Supplementary Appendix Table C.1](#)). We use the U.S. price level in 2007 and the average exchange rate between pounds sterling and U.S. dollars in that year. Using the average exchange rate over the full sample would make the dollar changes for the U.K. some 10% lower but would not affect the fact that the estimated values in each column are of the same order of magnitude across the two countries.

useful later when we quantify the contribution of different transmission mechanisms. In Table we convert the IRFs into an equivalent dollar change in expenditure over the four-year forecast period.¹² This can be seen as the overall dollar adjustment in the short-run as a result of the change in monetary policy.¹³ The overall effects on non-durable and durable expenditure in the

Table 1: Cumulative changes over four years in US dollars

5.4 Figure 5: U.K. demographic robustness and Figure 6: U.S. demographic robustness

Read: Middle-aged mortgagors respond strongly, while middle-aged outright owners do not. The pattern holds in both countries.

Economic message. The result is not simply an age or life-cycle effect.



FIGURE 5
Dynamic effects of a 25 bp anticipated interest rate cut on non-durable consumption (top) and durable expenditure (bottom) for mortgagors born after 1949 (left column), mortgagors born between 1935 and 1949 (middle column), and outright owners born before 1935 (right column). U.S. data: CE

(a) Figure 5: U.K. demographic robustness

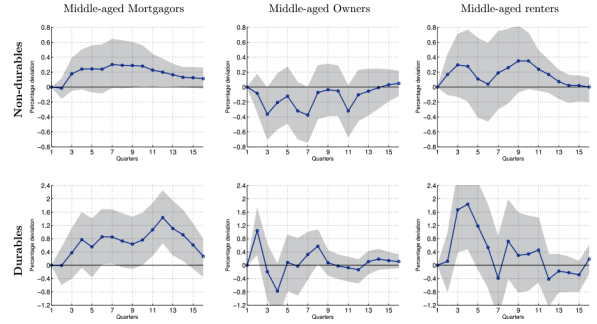


FIGURE 6
Dynamic effects of a 25 bp anticipated interest rate cut on non-durable consumption (top) and durable expenditure (bottom) for mortgagors born after 1949 (left column), mortgagors born between 1935 and 1949 (middle column), and outright owners born before 1935 (right column). U.S. data: CE

(b) Figure 6: U.S. demographic robustness

5.5 Figure 7: U.K. likely vs. unlikely mortgagors and Figure 8: U.S. likely vs. unlikely mortgagors

Read: Propensity-score likely mortgagors have stronger responses than unlikely mortgagors, especially for durables.

Economic message. Compositional changes in actual tenure status are unlikely to drive the results.

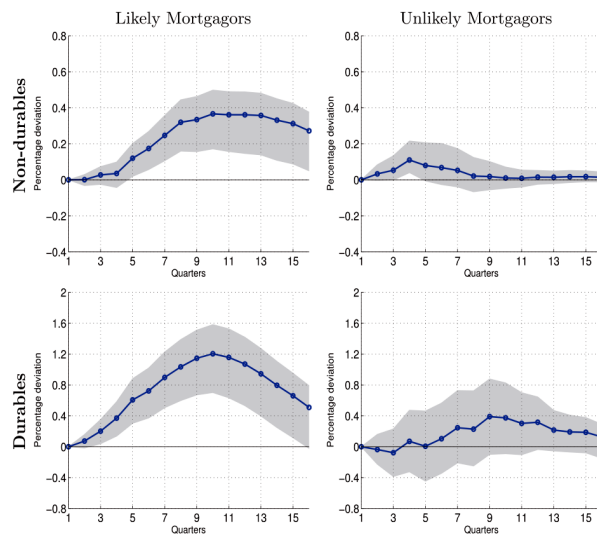


FIGURE 7

U.K. results for non-durable and durable expenditures for "likely" and "unlikely" mortgagors, excluding renters. Groups computed following Attanasio *et al.* (2002) propensity score approach using a fixed probability threshold. Gr

(a) Figure 7: U.K. likely/unlikely mortgagors

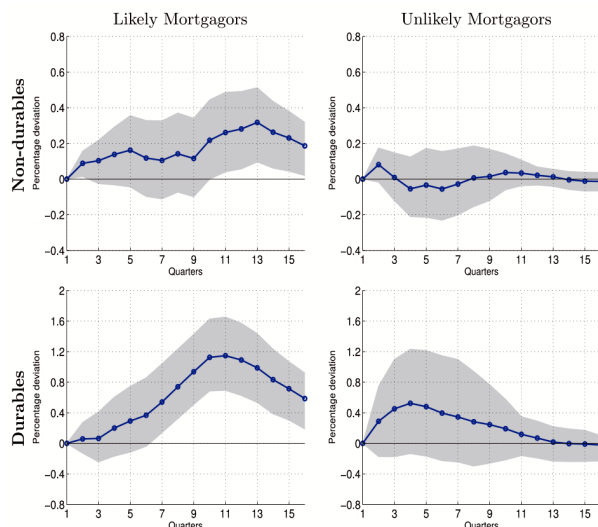


FIGURE 8

U.S. results for non-durable and durable expenditures for "likely" and "unlikely" mortgagors, excluding renters. Group

(b) Figure 8: U.S. likely/unlikely mortgagors

5.6 Figure 9: Gross and net income responses and Table 2: Household financial and housing wealth

Read: Income rises for all groups, but spending rises mainly for mortgagors and renters. Table 2 shows mortgagors hold housing wealth but little liquid wealth.

Economic message. General-equilibrium income effects interact with high-MPC balance-sheet positions.

5.7 Figure 10: Wealthy hand-to-mouth mortgagors

Read: Roughly 40–50% of mortgagors are wealthy hand-to-mouth. Most WHTM households have mortgages.

Economic message. This explains how mortgagors can be asset-rich but liquidity-poor.

5.8 Figure 11: Mortgage and rental payments

Read: Mortgage payments fall, especially in the U.K.; rental payments rise. The mortgage payment windfall is too small to explain the full spending response.

Economic message. The cash-flow channel is active but not sufficient.

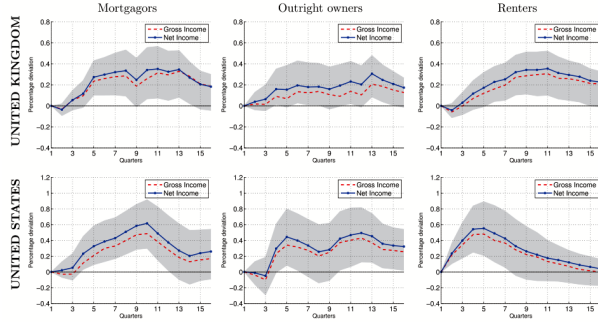


FIGURE 9

Dynamic effects of a 25 bp unanticipated interest rate cut on net income (blue) and gross income (red). Mortgagors (left), outright owners (centre), and renters (right). Gray areas are bootstrapped 90% confidence bands for net income

captionFigure 9: Net and gross income responses

TABLE 2
Household financial and housing wealth for the U.K. and U.S.

	U.K. BHPS: 1995		U.S. SCF: 1995	
	Median	[p25, p75]	Median	[p25, p75]
Net financial wealth				
Outright owners	2,000	[0, 10,000]	3,505	[165, 18,455]
Mortgagors	150	[−753, 3,450]	1,505	[−695, 8,705]
Renters	0	[−25, 559]	5	[−545, 2,005]
Net housing wealth				
Outright owners	59,000	[42,000, 80,000]	86,000	[52,000, 142,000]
Mortgagors	28,000	[10,000, 51,750]	43,500	[20,000, 86,000]
Renters	0	0	0	0
	U.K. BHPS: 2000		U.S. SCF: 2001	
	Median	[p25, p75]	Median	[p25, p75]
Net financial wealth				
Outright owners	3,300	[0, 15,000]	8,305	[1,005, 40,910]
Mortgagors	375	[−1,600, 6,400]	4,605	[105, 22,701]
Renters	0	[−68, 1,000]	145	[−255, 2,425]
Net housing wealth				
Outright owners	70,000	[47,000, 110,000]	111,500	[65,000, 203,500]
Mortgagors	36,000	[16,000, 70,000]	62,000	[25,000, 126,000]
Renters	0	0	0	0
	U.K. BHPS: 2005		U.S. SCF: 2004	
	Median	[p25, p75]	Median	[p25, p75]
Net financial wealth				
Outright owners	3,050	[0, 17,034]	5,580	[605, 56,805]
Mortgagors	0	[−3,250, 5,000]	2,500	[−95, 25,505]
Renters	0	[−455, 500]	70	[−345, 2,305]
Net housing wealth				
Outright owners	150,000	[100,000, 220,000]	145,000	[76,000, 265,000]
Mortgagors	97,000	[56,250, 152,000]	77,000	[32,000, 168,000]
Renters	0	0	0	0

Notes: *Net financial wealth* (NLW): (1) from the BHPS, the value of savings and investments net of outstanding net mortgage debt; (2) from the SCF, the value of checking, saving and MM accounts, directly held mutual funds, soci bonds and t-bills, net of outstanding unsecured debt. *Net housing wealth* is the household's estimate of the property value net of any outstanding mortgage and home equity lines of credit. Trimmed at the top 5 percentiles of the NLW distributi

Table 2: Household financial and housing wealth

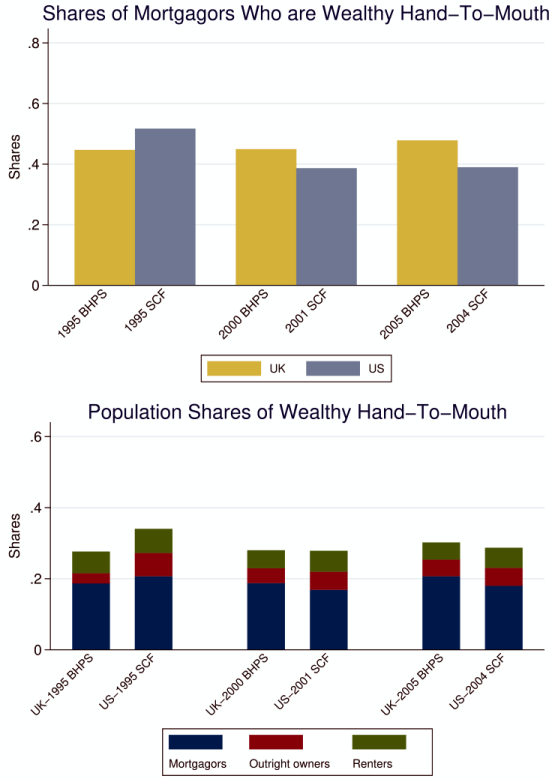


FIGURE 10

Figure 10: Shares of WHTM mortgagors

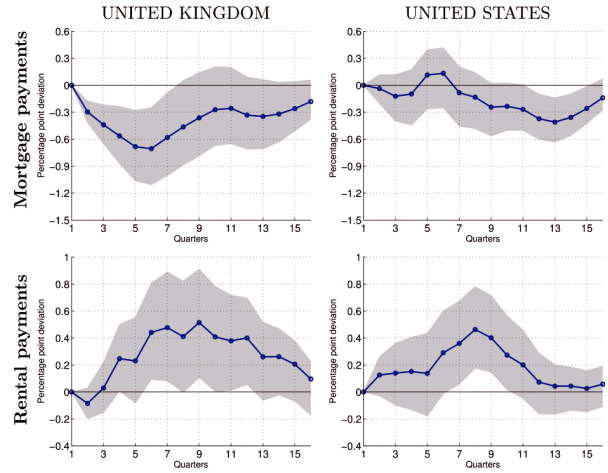


FIGURE 11

Figure 11: Mortgage and rental payments

5.9 Figure 12: CPI inflation response and Figure 13: Investment, government spending, and net exports

Read: Inflation rises only modestly and with delay. Investment responds strongly; government spending does not; net exports fall.

Economic message. Domestic private demand is the main source of the income increase.

own countries. This suggests that the income effect (on expenditure) dominates the expenditure switching effect and thus net exports make a negative contribution to aggregate demand when monetary policy is expansionary.

In summary, the boost to household income following a cut in interest rates reflects increased demand from the domestic private sector rather than expenditure by the government or the rest of the world. Interestingly, this is consistent with the theoretical results in the supplementary no

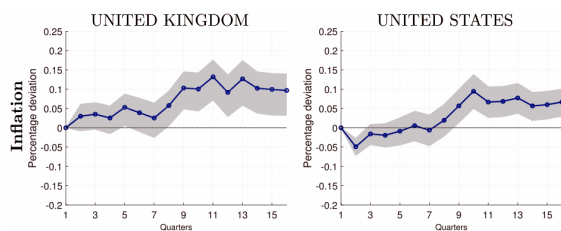


FIGURE 12
Dynamic effects of a 25 bp unanticipated interest rate cut on CPI inflation.

(a) Figure 12: CPI inflation

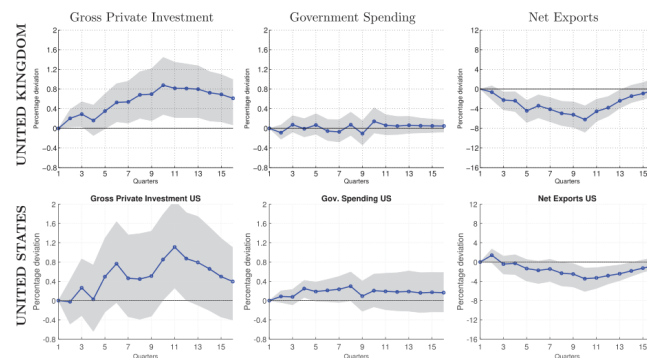


FIGURE 13
Dynamic effects of a 25 bp unanticipated interest rate cut on gross private investment (first column), government

(b) Figure 13: Income components

6 Short summary

- Monetary policy affects household spending differently by housing tenure.
- Mortgagors drive the aggregate consumption response to an interest-rate cut.
- Outright owners do not spend much more, even though income rises.
- Renters respond because they are poor hand-to-mouth; mortgagors respond because many are wealthy hand-to-mouth.
- Direct mortgage-payment windfalls are too small to explain the full response.
- The evidence supports HANK-style transmission through general-equilibrium income effects and MPC heterogeneity.

7 Conclusion

7.1 Core conclusion

Cloyne, Ferreira, and Surico show that conventional monetary policy is transmitted through heterogeneous household balance sheets. Mortgagors are central not only because they owe debt, but because many have illiquid housing wealth and little liquid wealth.

7.2 Mechanism

A rate cut raises aggregate demand and household income. High-MPC households—renters and liquidity-poor mortgagors—turn that income into expenditure. Outright owners receive income gains but spend little of them.

7.3 What the paper rules out

The paper rules out pure cash-flow, EIS, inflation-redistribution, financial wealth, fiscal, and external-demand channels as complete explanations.

7.4 Policy implications

Central banks should account for liquid wealth, illiquid wealth, and debt jointly. Mortgage-market design matters, but direct pass-through is only one part of the mechanism.

7.5 Limitations

Housing tenure is a proxy; the data are repeated cross sections; high-frequency wealth and mortgage-contract data are unavailable; the sample ends before the zero lower bound.

7.6 Takeaway

Monetary policy works through the households that are likely to spend extra income. Household liquidity and portfolio composition are therefore first-order determinants of monetary transmission.